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Aim

To implement page frame with FIFO and LRU system and calculate the number of hits and misses.

Problem Statement

Consider a system with 3 page frames and the following reference string containing 20 page references. Calculate the total page faults and page hits using FIFO and LRU page replacement policies.

```
# ref_str = '7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1,
# ref_arr = ref_str.split(',')
# ref_arr = [int(x) for x in ref_arr]
ref_arr = [7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7
frame_no = 3
```

1. FIFO Page Replacement

```
page = [-1, -1, -1]
page_ptr = miss = hit = 0
for i in range(len(ref_arr)):
    if page_ptr == 3: # Loop around condition
        page_ptr = 0
    if ref_arr[i] in page: # if exists in page then a hit
        print(i, 'H', end=" ")
        hit+=1
    else:
        page[page_ptr] = ref_arr[i]
        print(i, 'F', end=" ")
        miss+=1
        page_ptr+=1
    print(page)

print("Total no of Miss/Faults = ", miss)
print("Total no of Hit/Non Faults = ", hit)
print("Miss/Hit Ratio: ", miss/hit)
```

```
0 F [7, -1, -1]
1 F [7, 0, -1]
2 F [7, 0, 1]
3 F [2, 0, 1]
4 H [2, 0, 1]
5 F [2, 3, 1]
6 F [2, 3, 0]
```

```
7 F [4, 3, 0]
8 F [4, 2, 0]
9 F [4, 2, 3]
10 F [0, 2, 3]
11 H [0, 2, 3]
12 H [0, 2, 3]
13 F [0, 1, 3]
14 F [0, 1, 2]
15 H [0, 1, 2]
16 H [0, 1, 2]
17 F [7, 1, 2]
18 F [7, 0, 2]
19 F [7, 0, 1]
Total no of Miss/Faults =  15
Total no of Hit/Non Faults =  5
Miss/Hit Ratio:  3.0
```

2. LRU Page Replacement.

```
page = [-1, -1, -1]
page_use = [-1, -1, -1]
miss = 0
hit = 0

for i in range(len(ref_arr)):
    if ref_arr[i] in page: # if exists in page then a hit
        page_use[page.index(ref_arr[i])] = i
        print(i, 'H', end=" ")
        hit+=1
    else:
        replace = page_use.index(min(page_use))
        page[replace] = ref_arr[i]
        page_use[replace] = i
        print(i, 'F', end=" ")
        miss+=1
    print(page, page_use)

print("Total no of Miss/Faults = ", miss)
print("Total no of Hit/Non Faults = ", hit)
print("Miss/Hit Ratio: ", miss/hit)
```

```
0 F [7, -1, -1] [0, -1, -1]
1 F [7, 0, -1] [0, 1, -1]
2 F [7, 0, 1] [0, 1, 2]
3 F [2, 0, 1] [3, 1, 2]
4 H [2, 0, 1] [3, 4, 2]
5 F [2, 0, 3] [3, 4, 5]
6 H [2, 0, 3] [3, 6, 5]
7 F [4, 0, 3] [7, 6, 5]
8 F [4, 0, 2] [7, 6, 8]
9 F [4, 3, 2] [7, 9, 8]
10 F [0, 3, 2] [10, 9, 8]
11 H [0, 3, 2] [10, 11, 8]
12 H [0, 3, 2] [10, 11, 12]
13 F [1, 3, 2] [13, 11, 12]
```

```
14 H [1, 3, 2] [13, 11, 14]
15 F [1, 0, 2] [13, 15, 14]
16 H [1, 0, 2] [16, 15, 14]
17 F [1, 0, 7] [16, 15, 17]
18 H [1, 0, 7] [16, 18, 17]
19 H [1, 0, 7] [19, 18, 17]
Total no of Miss/Faults =  12
Total no of Hit/Non Faults =  8
Miss/Hit Ratio:  1.5
```

Conclusion

A 3 frame page was implemented in python with FIFO and LRU type and tested to determine the ratio of misses to hits.